A3-PS031-TC557

Offline ASPA-Based BGP Export Policy Checker

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Problem Description

Explainable, offline verification of BGP AS_PATHs using ASPA

We propose an offline tool that checks BGP AS_PATHs against ASPA provider-authorization information, following the verification algorithm revision supplied by the hackathon mentors.

The problem

BGP routing depends on inter-domain path information. Misconfigured or malicious routing behaviour can cause route leaks and path-security concerns.

The challenge

A checker must follow the specified ASPA algorithm and reason carefully when records or policy context are missing.

Our goal

Return **Valid**, **Invalid** or **Unknown** with a clear, evidence-based explanation for every result.

Scope: offline verification with synthetic ASPA records, AS_PATHs, neighbour relationships and controlled test cases. No router implementation and no live route blocking.

Key deliverables: verification engine • 20+ reviewed cases • expected-result tests • explanation / audit output • demonstration and report

Background

Concepts needed to understand the proposed checker

BGP

Border Gateway Protocol exchanges reachability information between autonomous systems (ASes).

AS_PATH

An ordered sequence of AS numbers attached to a route announcement; the verifier checks it with the specified procedure.

RPKI / ASPA

RPKI supports routing-security data. An ASPA record names a customer AS and its authorized provider ASes.

Provider relationship

Verification uses provider-authorization data to evaluate each relevant relationship along the path.

Three outcomes

Valid, Invalid and Unknown stay distinct, especially when required data is unavailable.

Offline testbed

Synthetic records and reproducible inputs let us test edge cases without touching production routing.

Implementation rule: follow the exact algorithm revision and expected-result vectors agreed with the hackathon mentors; do not substitute a different ASPA procedure.

Literature & RFC Review

Most relevant standards and references, studied before coding

Reference	Why it matters to PS-031	How we will use it
draft-zhang-sidrops-aspa-egress (named in the problem statement)	Defines ASPA-based checking for BGP route export, the direction this challenge follows.	Primary source, together with the mentor-pinned revision.
draft-ietf-sidrops-aspa-verification BGP AS_PATH Verification Based on ASPA Objects	Defines ASPA path-verification procedures and outcomes.	Background reading; the mentor-pinned revision is the authority.
RFC 6811 BGP Prefix Origin Validation	RPKI origin validation: related context, not a replacement for ASPA path verification.	Explain origin validation vs path verification.
NIST BRIO	Experimentation framework for studying BGP security mechanisms.	Background for controlled, reproducible testing.
PS-031 statement and mentor test vectors	Defines deliverables, case coverage and expected behaviour.	Acceptance checklist for the implementation.

Note: confirm the exact draft revision and any mentor-supplied algorithm changes before implementation. The agreed revision and test vectors determine project correctness.

Proposed Solution

A small, testable verification engine with explicit explanations

Input layer

JSON/YAML fixtures for synthetic ASPA records, AS_PATHs, neighbour relationships, direction and policy context.

Core engine

Plain Python modules implementing the mentor-specified algorithm with deterministic results and explicit handling of missing data.

Evidence layer

Every outcome lists the checks performed, the relevant evidence, and a reason matching the actual verification logic.

Proposed stack: Python • pytest • JSON/YAML • optional simple demo UI. No ML model is required.

Optimization Proposed

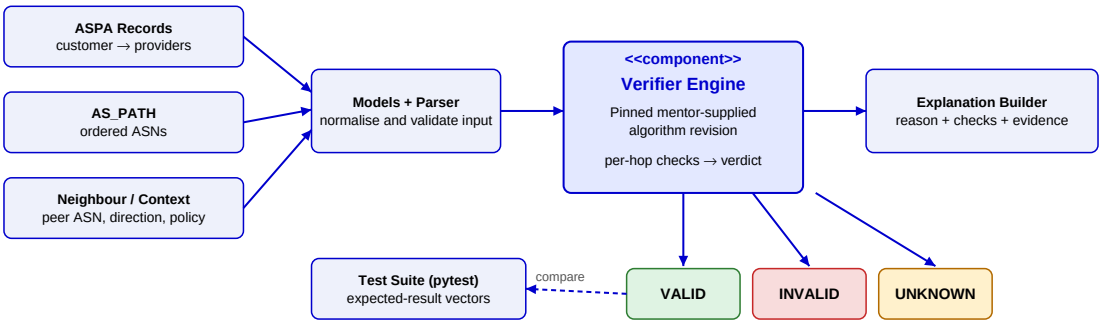
Improvements beyond a minimal pass/fail script

Area	Basic approach	Proposed improvement	How we will check it
Result handling	Single status output	Separate Valid / Invalid / Unknown semantics	Compare all outputs with expected vectors
Explanations	Generic message	Per-hop checks and evidence for each result	Review explanation against engine trace
Test coverage	A few happy-path examples	20+ reviewed valid, invalid and incomplete-data cases	Automated pytest suite
Reproducibility	Manual inputs	Versioned fixtures and machine-readable output	Repeatable test run and saved report
Usability	Terminal output only	Readable AS_PATH view and result summary	Demo using representative cases

We prioritise correctness and explainability before interface polish or performance optimisation.

Solution Architecture & Design

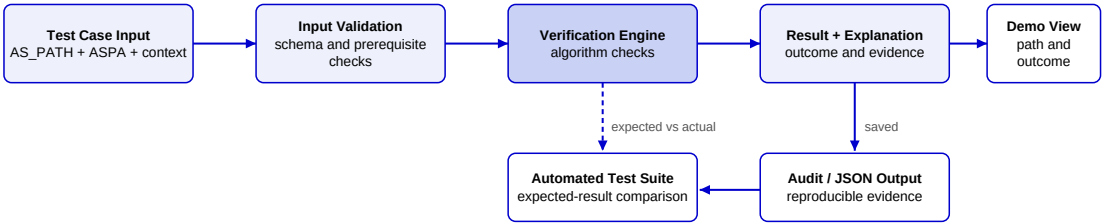
UML component diagram of the offline verifier



All outputs include an explanation and an auditable test result. The verifier follows the specified procedure without silently substituting another draft version.

Data Flow Diagram

How information moves from a test case to an auditable result



Data flow: fixtures are loaded and validated, passed to the verifier, then converted into a result plus explanation. The test suite checks that result against the agreed expected output; the demo view only presents the engine's actual output.

Deployment Timeline

Proposed 30-day plan (adjust to the official hackathon schedule)

Days	Milestone	Activities	Exit criterion
1–4	Understand specification	Study BGP/AS_PATH, ASPA and the mentor-pinned draft.	Algorithm flowchart and glossary reviewed by mentor
5–10	Build data model	Define ASPA records, path input, neighbour/context models, input validation.	Structured test fixtures load consistently
11–16	Implement verifier	Implement required verification procedures and outcome semantics.	Core engine runs end-to-end on initial cases
17–21	Test correctness	Create 20+ reviewed cases; cover missing records and policy context.	Automated expected-result tests pass
22–26	Explanations + demo	Add reason/evidence output and optional AS_PATH view.	Results can be explained in a demo
27–30	Validate + present	Review edge cases, document limitations, rehearse demo, write report.	Reproducible test report and final presentation

Bibliography I

[1] J. Zhang et al., *ASPA-based AS_PATH Verification for BGP Export*, Internet-Draft draft-zhang-sidrops-aspa-egress-05 (work in progress), July 2026. Named in the PS-031 statement; revision to be confirmed with the mentor.
https://datatracker.ietf.org/doc/draft-zhang-sidrops-aspa-egress/05/

[2] IETF SIDROPS WG, *BGP AS_PATH Verification Based on Autonomous System Provider Authorization (ASPA) Objects*, Internet-Draft draft-ietf-sidrops-aspa-verification (latest version seen: -28, August 2026).
https://datatracker.ietf.org/doc/draft-ietf-sidrops-aspa-verification/

[3] P. Mohapatra et al., *BGP Prefix Origin Validation*, RFC 6811, January 2013. Background on origin validation; not a substitute for ASPA path verification.
https://www.rfc-editor.org/rfc/rfc6811.html

[4] NIST, *NIST BGP RPKI IO (BRIO)*, open-source test tool for BGP/RPKI security mechanisms including ASPA, announced in “NIST Releases Test Tools to Accelerate Adoption of Emerging Route Leak Mitigation Standards”.
https://www.nist.gov/news-events/news/nist-releases-test-tools-accelerate-adoption-emerging-route-leak-mitigation

[5] AIORI-3, *PS-031: Offline ASPA-Based BGP Export Policy Checker*, official problem statement; hackathon page:
https://portal.aiori.in/aiori-3/hackathon/

Thank You for Your Attention!

Building an explainable and reproducible ASPA verification tool

Team TrustZero

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